

MASTER'S THESIS · DATA SCIENCE

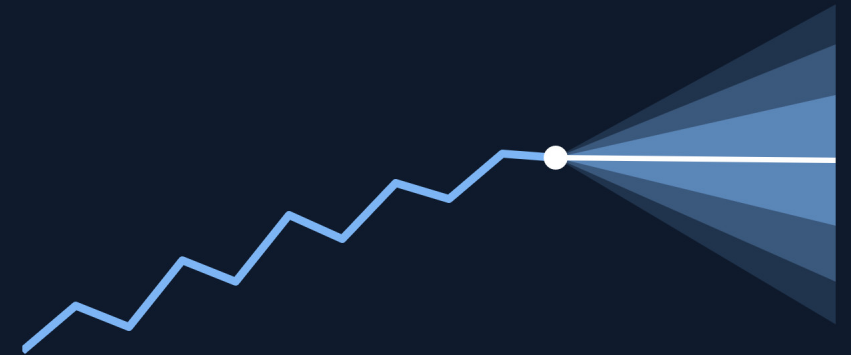
RiskLens

Risk-aware stock forecasting and trading decision support

Calibrated volatility forecasts and prediction ranges, with a signal layer that is tested honestly.

Nazmul Farooquee · September 2026

Academic prototype, not financial advice.



THE PROBLEM

Investors see prices and point forecasts, never the risk around them

THE DAILY QUESTION

How risky is the market right now,
what range should I expect tomorrow,
and is holding the informed choice?

USER

A non-professional investor who checks one asset for a few minutes a day.

GAP

Free tools show charts and point predictions. The uncertainty is invisible.

CONSEQUENCE

Entering calm markets by mistake and exiting in panic: emotion-driven timing.

APPROACH

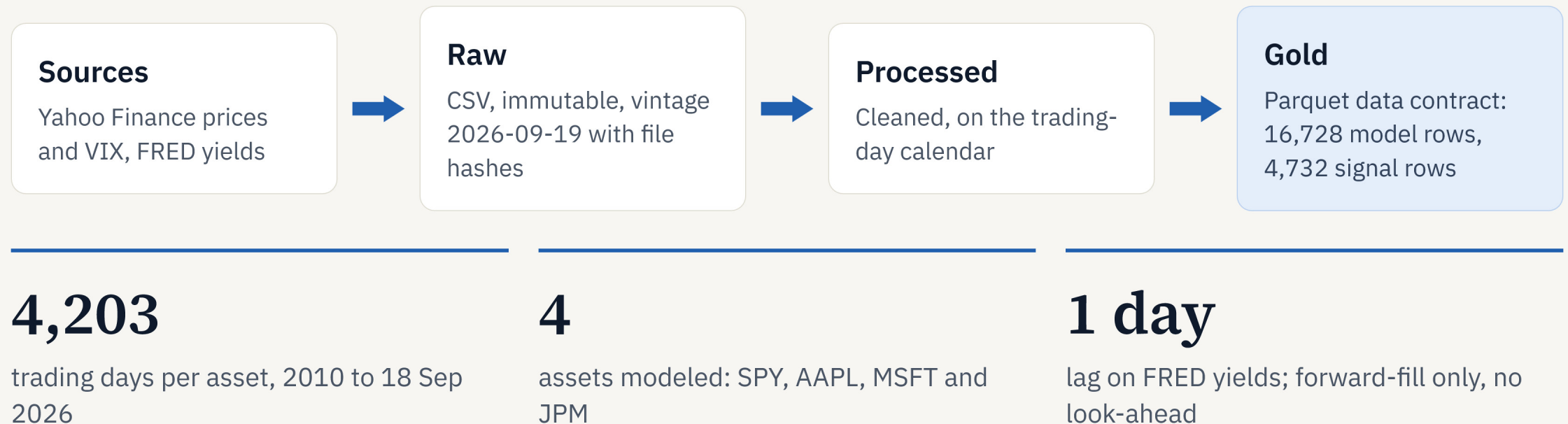
Risk first: two forecasts and one transparent rule

Task	Model	Baseline to beat	Primary metric
Risk	GARCH(1,1) with Student-t errors	21-day rolling volatility	QLIKE, 95% interval coverage
Return	ARIMA and ARIMAX	Naive zero forecast	RMSE, directional accuracy
Decision	Score rule and volatility filter	Buy-and-hold of the same asset	Sharpe and drawdown after costs

Core: a validated risk system with calibrated ranges

Conditional: signals, promoted only if proven out of sample

Open data through three layers, frozen as one vintage



The test set stays sealed until the model is fixed



Train 2010 to 2021: fit models, select orders

Validation 2022 to 2023: fix thresholds

Test 2024 to 18 Sep 2026: opened once

Walk-forward

Expanding window, refit every 21 trading days

Thresholds

Chosen on validation data only, then frozen

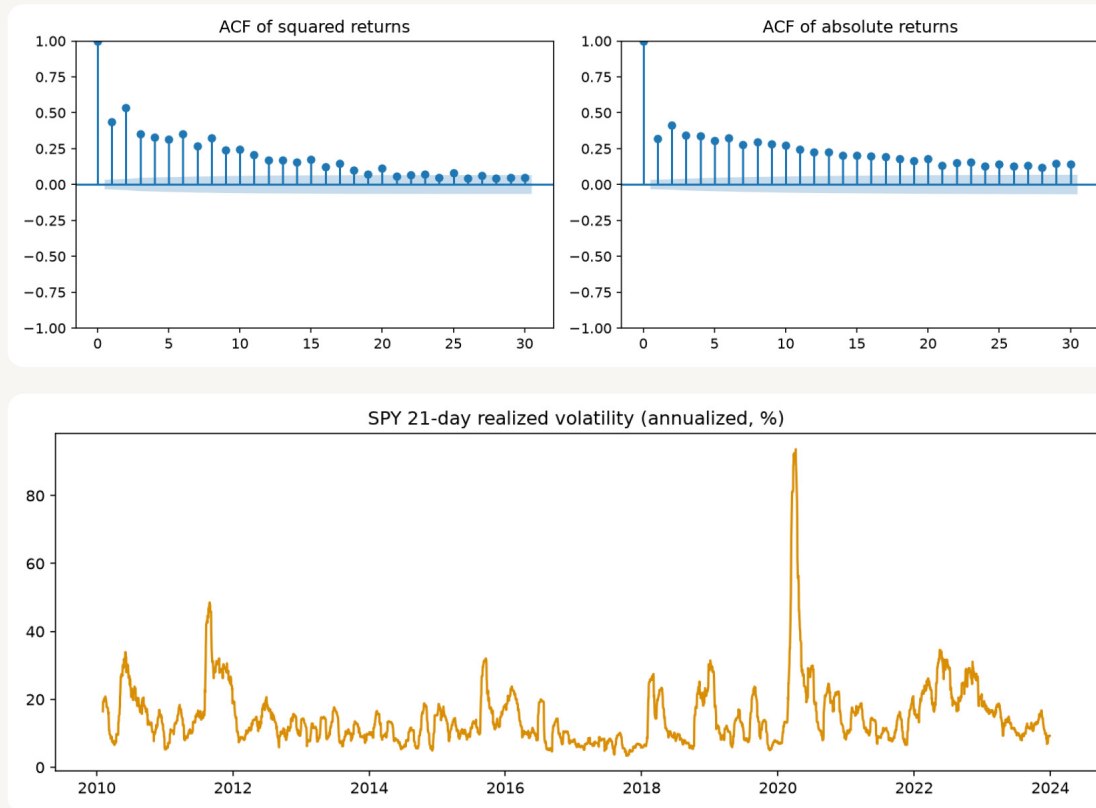
Execution

A signal at the close of day t trades at the close of $t+1$

Pre-registered

Models and rules committed and tagged before the test

Volatility is predictable; returns are not



0.46

21-day volatility vs next-day absolute return

11.6

excess kurtosis: heavy tails, so Student-t errors

-0.12

leverage effect: bad days precede bigger moves

47%

of the largest 1% of moves fall in 2020

Calibrated risk, no edge in returns

Volatility: QLIKE, lower is better

Model	QLIKE	95% coverage
Rolling 21-day	-7.948	93.6%
EWMA	-7.974	94.2%
GARCH-t	-7.946	95.2%
GJR-GARCH-t	-7.944	94.2%

Next-day return: RMSE, lower is better

Model	RMSE	vs naive
Naive zero	0.01229	baseline
ARIMA(2,0,0)	0.01247	worse, p 0.02
ARIMAX + VIX	0.01246	worse, p 0.01

Outcome: no volatility model beat the rolling baseline significantly, and every return model was significantly worse than forecasting zero.

SEALED TEST, 2024 TO 18 SEP 2026

The risk forecasts hold up on four assets

4 of 4

assets where GARCH beats rolling volatility on QLIKE ($p = 0.002$ to 0.010)

94-95%

GARCH 95% interval coverage on every asset (Kupiec p above 0.3)

4 of 4

rolling-volatility intervals rejected as miscalibrated (coverage 91.9% to 93.0%)

SPY 95.2%

AAPL 94.1%

MSFT 94.6%

JPM 94.9%

SPY fan-chart bands: 50% covered 49.5%, 80% covered 78.7%, 95% covered 95.3%.

No validated edge on any asset

Asset	Buy-and-hold	Score rule A	Filter B	Reading
SPY	1.27	0.73	0.17	Neither beats buy-and-hold
AAPL	0.90	0.95	0.07	A = buy-and-hold (1 trade)
MSFT	0.54	0.35	0.18	Neither beats buy-and-hold
JPM	1.33	1.01	0.70	Neither beats buy-and-hold

RETURNS

ARIMA never beats the naive forecast;
significantly worse for MSFT (p = 0.048)

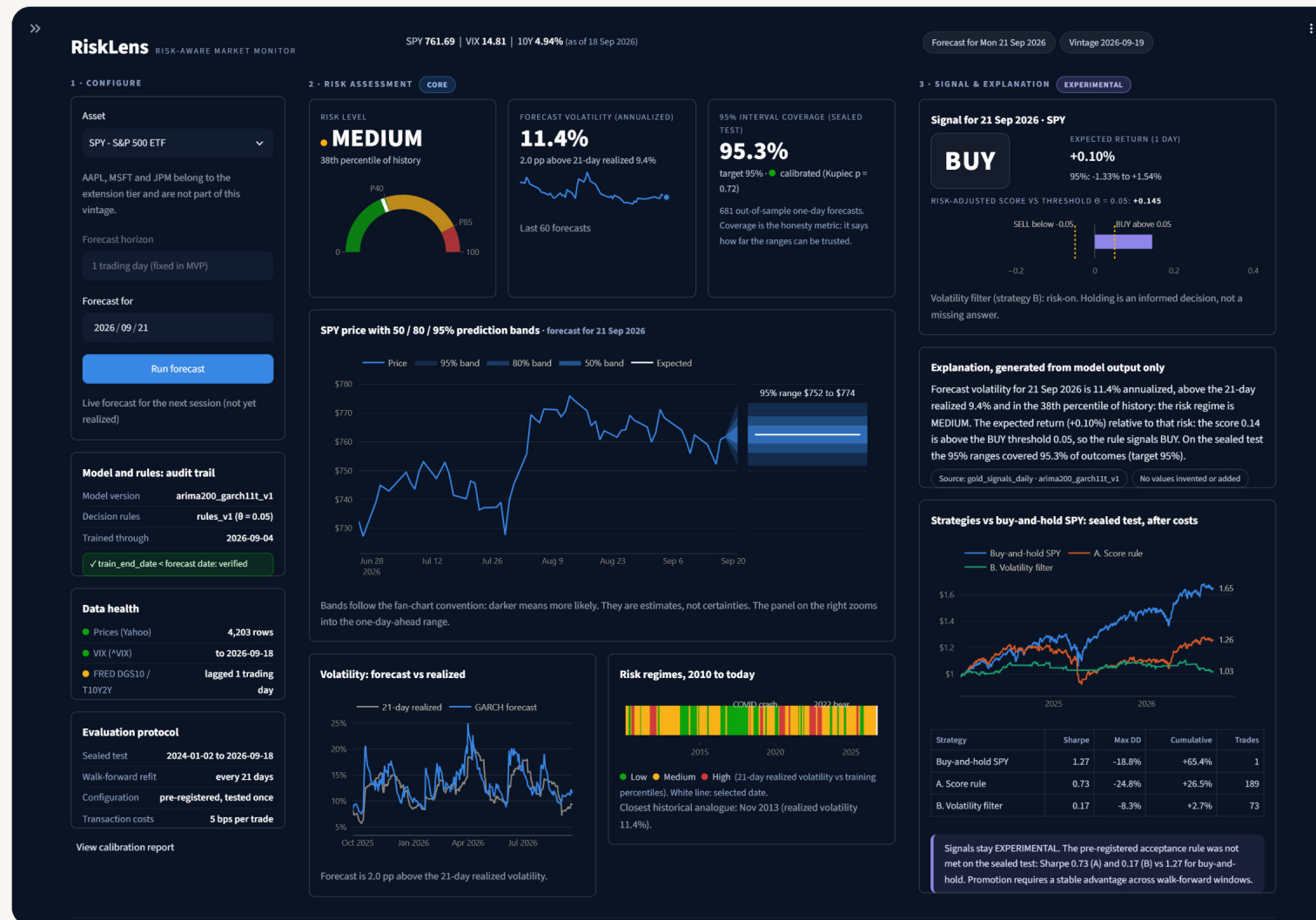
COSTS

SPY score rule: Sharpe 1.00 at 0 bps
falls to -0.07 at 20 bps

REGIME

2024 to 2026 was a strong bull
market, which favors holding

Signals stay labeled EXPERIMENTAL on every asset



THE PRODUCT

RiskLens dashboard

1 • Configure

Asset (SPY, AAPL, MSFT, JPM), date, audit trail, data health

2 • Risk assessment, core

Gauge, forecast vs realized volatility, fan chart, regimes, coverage

3 • Signal, experimental

Score vs threshold, grounded explanation, backtest vs buy-and-hold

LIVE APP

risklensspy.streamlit.app

Built so that every number can be audited



Pre-registered

Models and thresholds are tagged before the test. The test runner refuses to run if frozen files changed.



No look-ahead

Every signal row asserts training end before forecast date; features use past data only, and tests prove it.



Grounded explanations

The text restates numbers shown on screen and never adds causes or outside facts.



Reproducible

Frozen data vintage with file hashes, a locked environment, 102 automated tests, one command per stage.

CONCLUSIONS

A validated risk system and an honest negative result

Delivered

- Calibrated 50, 80 and 95% ranges on four assets, tested out of sample
- GARCH beats rolling volatility on all four assets
- Pipeline, audit trail and dashboard
- Live-data refresh with a backup source, kept outside the sealed test

Limits

- Four large-cap US assets, no other markets
- Daily data only, squared-return proxy
- Test window is a single bull-market regime
- Yahoo is unofficial; frozen vintage and backup source mitigate it

Next

- More assets and sectors, each pre-registered
- Intraday realized volatility
- Regime-aware volatility models

Questions?